

I've now read all six sources. Here's the populated fact list.

## The fact list

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### Tier 1 — the spine of the story (use these in the 3D)

1. **Chassis:** *Saccharomyces cerevisiae* BY4741 — GRAS, fully sequenced, big toolkit, strong homologous recombination
2. **Goal:** accumulate **triacylglycerols (TAGs)** in lipid droplets, then release them **at cooking temperature**
3. **The switch:** a synthetic **RNA thermometer in the 5' UTR of BGL2** — a hairpin that stays closed at growth temperature and unfolds on heating
4. **The release:** **BGL2** = endo- $\beta$ -1,3-glucanase; degrades the **cell wall** until it can't hold internal hydrostatic pressure  $\rightarrow$  lysis  $\rightarrow$  lipid release
5. **Why it matters:** fat is the sensory experience of meat; alternative proteins lack intramuscular marbling
6. **Novelty:** to their knowledge, **no synthetic RNAt has been shown for temperature-controlled translation in *S. cerevisiae*** — bacterial RNAts mask the Shine-Dalgarno site, but yeast uses cap-scanning, so this is an adaptation, not a port

### Tier 2 — the five engineered genes (one construct, five transcription units)

| Gene          | Role                                                                                                       | Promoter strength                              |
|---------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| <b>ACC1**</b> | acetyl-CoA $\rightarrow$ malonyl-CoA; <b>double mutant Ser659Ala + Ser1157Ala</b> relieves Snf1 repression | medium                                         |
| <b>DGA1</b>   | final acylation DAG $\rightarrow$ TAG                                                                      | strong                                         |
| <b>OLE1</b>   | $\Delta 9$ desaturase; saturated $\rightarrow$ monounsaturated (healthier)                                 | weak-medium                                    |
| <b>TPS1</b>   | trehalose synthesis; freeze/dry/heat tolerance                                                             | weak-medium                                    |
| <b>BGL2</b>   | the lysis enzyme, <b>RNAt-gated</b>                                                                        | cpTEF_6 core promoter, <b>69 bp, no 5' UTR</b> |

### Tier 3 — knockouts (what's switched off)

- **$\beta$ -oxidation:** POX1, PXA1, FAA2 — stop burning the stored fat
- **Lipases:** TGL3/4/5 — stop mobilising TAG out of droplets
- **GUT2** — stop glycerol-3-P being pulled back to glycolysis

### Tier 4 — real numbers you can show

- **+31.2%** neutral lipid in PXA1 knockout vs BY4741 (Nile Red, n=6) — **your one real wet-lab result**
- **$\approx 91.84$  mg** neutral lipid per g dry cell weight (vs  $\sim 70$  mg baseline)
- **BGL2 = 313 aa**;  $\sim 9.5$  aa/s  $\rightarrow \sim 33$  s per protein

- RNAt model: three designs at **Tm 44.5 / 39.0 / 36.5 °C; 45 °C lethal**; Sequence 3 (36.5 °C) wins
- Kinetic model: TAG plateaus at **~560 mM** at ~55 min, limited by **precursor exhaustion**, not enzymes
- ML1 (ThermoRank): RMSE **1.03** and **0.93** for MFE; **10.20** for Tm (weak)
- ML2 (ThermoCast): **~398,000** UTRs; UTR-LM best, **p ≈ 0.6**; RNABERT p = 0.296 ≈ no better than thermodynamics alone

#### Tier 5 — honest limitations (worth showing, judges reward it)

- **The final construct was never assembled** — OYC-dropout-suGFP wouldn't transform; epPCR never attempted; in-silico results promising only
- No Nile Red calibration curve, so lipid-per-dry-weight is an estimate
- Kinetic model borrows Km values from *E. coli*, *M. tuberculosis*, *R. norvegicus*
- ML Tm predictions fail outside training distribution (one was off by −21.6 °C)
- No antibiotic selection — URA3 auxotrophy instead, deliberately, for food-safety/FDA

## What I'd actually put in the 3D blueprint

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**The cell:** lipid droplets (the payload), cell wall (the barrier BGL2 attacks), ER with OLE1, with the four pathway labels ACC1\*\* → malonyl-CoA → FAS → acyl-CoA → DGA1 → TAG.

**The hero moment:** the RNAt hairpin unfolding → BGL2 translating → wall thinning → droplets releasing. That's your whole project in one animation, and it's the part nobody else will have.

**What I need from your wet-lab lead:** the correct burst temperature, and whether to use 50 °C, 60 °C, or the modelled 36.5 °C. Everything else above comes straight from your own pages, but that number is contradictory at the source and it's the single most prominent figure in the design.